Predictive Modeling - Sales Prediction

1. Introduction

This project focuses on predicting product sales based on advertising budgets. Advertising plays a major role in increasing sales for any business. In this task, we use machine learning to find the relationship between advertising spend and sales. The goal is to build an accurate model that can forecast future sales.

Data Moldel

Name: Advertising Dataset Rows: 200 Columns: TV, radio, newspaper, sales Target Variable: sales3. Program Code

2. Dataset Description

We used the Advertising Dataset for this project. The dataset contains 200 records. It has 4 columns: TV, Radio, Newspaper, and Sales. Here, TV, Radio, and Newspaper are the input features. Sales is the target variable that we want to predict. The data shows how much money was spent on each type of advertising and the resulting sales.

3. Data Visualization

First, we created a pairplot to understand the relationship between all variables. From the pairplot, we can see that TV advertising has a strong positive relationship with sales. Radio also shows a good relationship, but Newspaper has a weaker relationship with sales.

4. Model Building

For this predictive task, we selected the Random Forest Regressor algorithm. Random Forest is good because it can handle non-linear relationships and gives accurate results. We split the dataset into 80% for training and 20% for testing. The model was trained on the training data and then tested on the test data.

PROGRAM

SALES PREDICTION WITH RANDOM FOREST

```
# 1. Import all libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns
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from sklearn. model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_squared_error

# 2. Load Dataset from internet
url = "https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv"
df = pd.read_csv(url)

# Remove the extra first column
df = df.iloc[:, 1:]

print("Dataset Loaded Successfully")
print(df.head())

# 3. Create Pairplot and save it
sns.pairplot(df)
plt.suptitle("Pairplot: TV vs Radio vs Newspaper vs Sales", y=1.02)
plt.savefig("pairplot.png", dpi=300, bbox_inches='tight')
plt.show()

# 4. Split data into X and y
# Note: column names are small letters
X = df[['TV', 'radio', 'newspaper']]
y = df['sales']

# Split into Train and Test
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 5. Create and Train Random Forest Model
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model = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
model.fit(X_train, y_train)

# 6. Predict on Test data
y_pred = model.predict(X_test)

# 7. Calculate Accuracy
r2 = r2_score(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

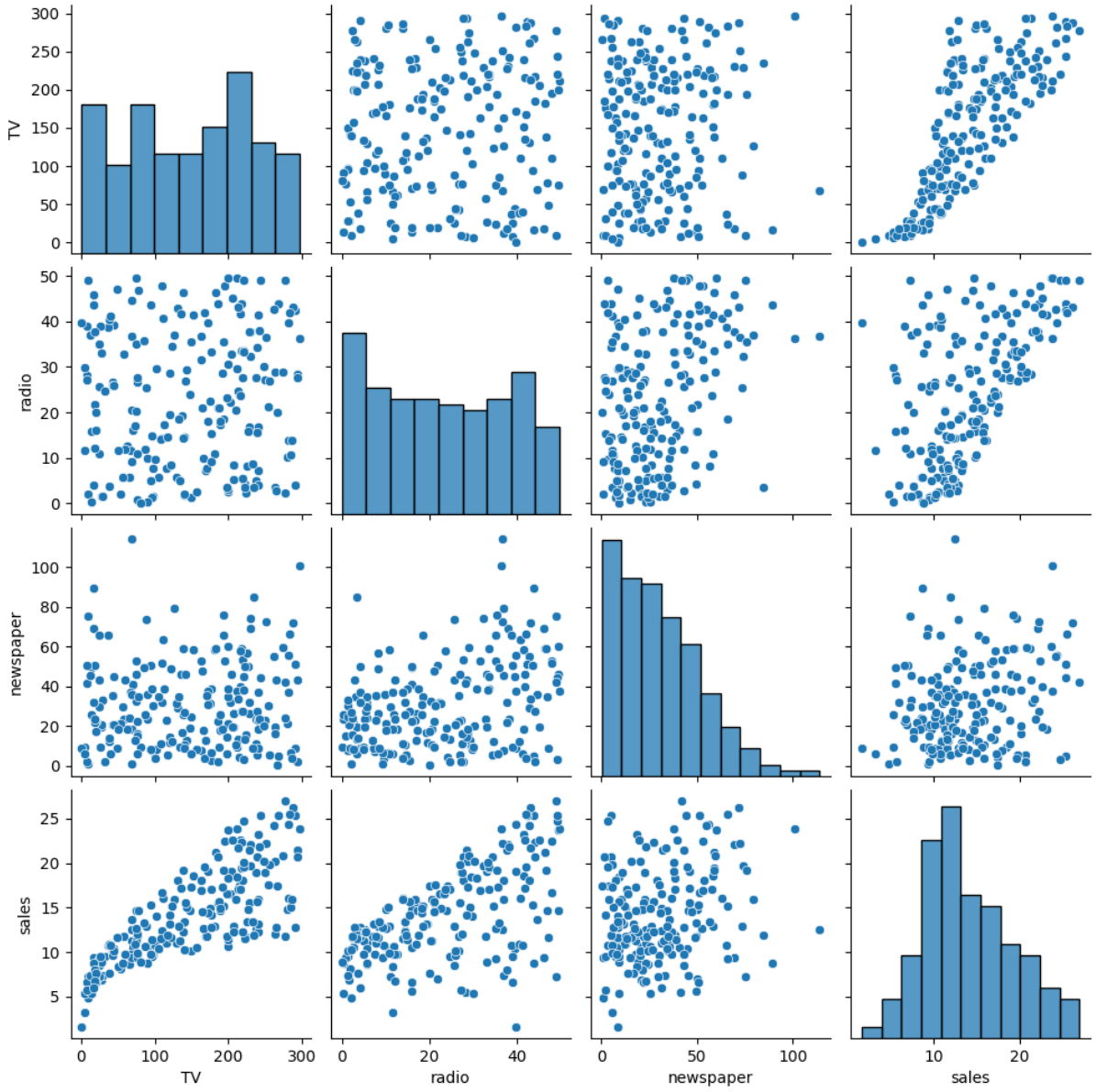
print("\n================ RESULTS ================")
print(f"R2 Score: {r2:.4f}")
print(f"RMSE: {rmse:.4f}")
print("=========================================")

# 8. Plot Actual vs Predicted and save it
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.8, color='royalblue', edgecolor='black')
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--', lw=2, label='Perfect Prediction')
plt.xlabel("Actual Sales", fontsize=12)
plt.ylabel("Predicted Sales", fontsize=12)
plt.title("Actual vs Predicted Sales", fontsize=14)
plt.legend()
plt.grid(True, alpha=0.2)
plt.savefig("result_graph.png", dpi=300, bbox_inches='tight')
plt.show()

print("\nDone! Check left side 'Files' for pairplot.png and result_graph.png")
```

Data Loaded

	TV	radio	newspaper	sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	9.3
3	151.5	41.3	58.5	18.5
4	180.8	10.8	58.4	12.9

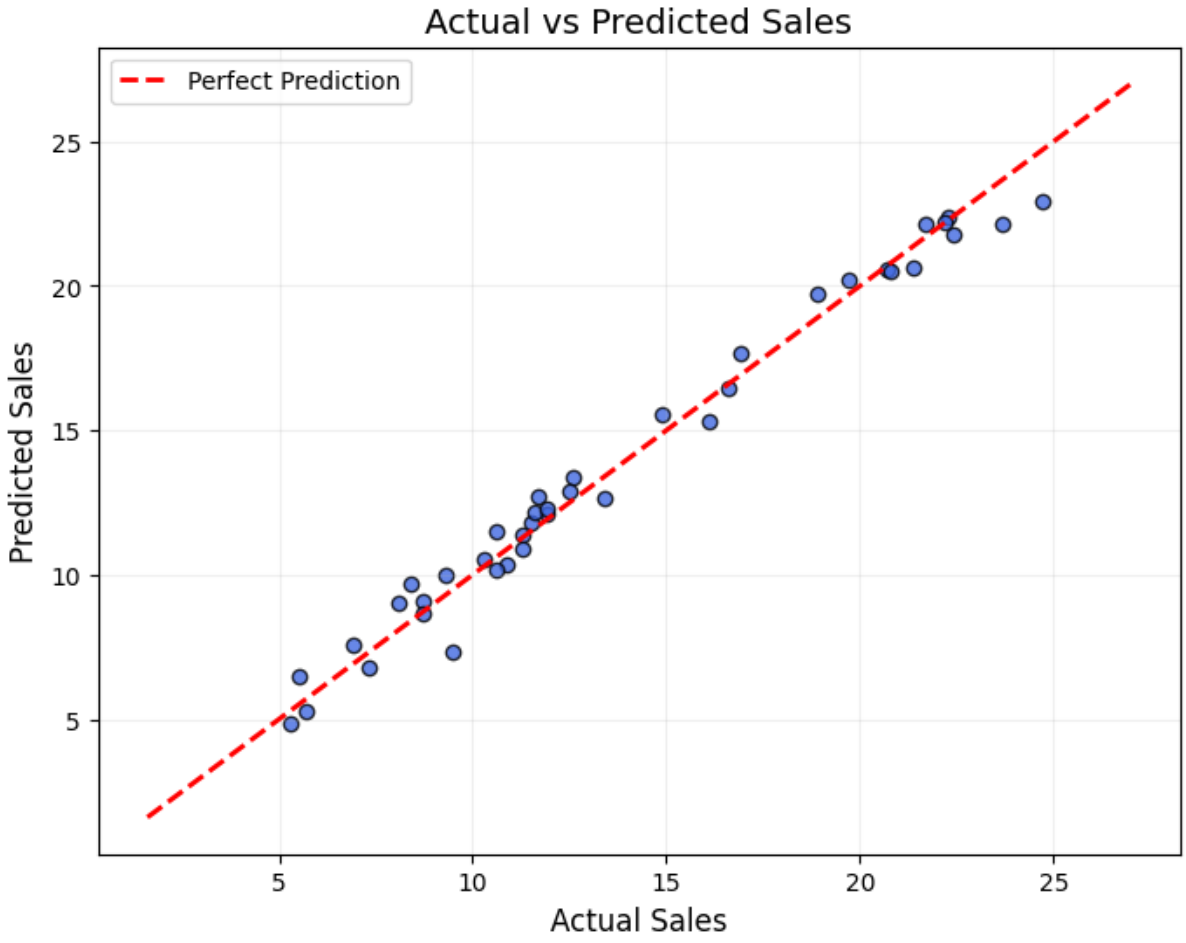


========== **RESULTS** ==========

R2 Score: 0.9813

RMSE: 0.7686

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5. Model Evaluation and Results

After training, we tested the model on new data. The model performed very well. R^2 Score: 0.9813 RMSE: 0.7686 An R^2 score of 0.9813 means the model can explain 98.13% of the variation in sales. The Actual vs Predicted graph also shows that most of the predicted points are very close to the actual values. This proves the model is highly accurate.

6. Conclusion

In conclusion, the Random Forest model is very effective for predicting sales from advertising data. The results show that TV advertising has the biggest impact on sales, followed by Radio. This model can help companies decide how to spend their advertising budget to get maximum sales**.**